KACPER SZYMANSKI

Software Engineer / Computer Science Student

Edinburgh, United Kingdom

kacszymansk@gmail.com ◆ **www.linkedin.com/in/kacper-szymanski-edi/** ◆ **github.com/aux1r**

EDUCATION

University of Edinburgh **Edinburgh, United Kingdom**

Bachelor of Engineering in Software Engineering Sep 2024 – Present

- **Courses**: Functional Programming, Object Oriented Programming, Computer Systems, Data Science, Algorithms & Data Structures, Software Engineering & Professional Practice

TECHNICAL SKILLS & LANGUAGES

- **Languages**: English (Native speaker), Polish (Native speaker)
- **Programming**: C, Rust, Python, Java, Javascript, SQL, Haskell, R, MATLAB, x86 Assembly
- **Frameworks & Libraries**: React, Vue, Express, Pandas, Matplotlib, Seaborn, Numpy
- **Tools**: Git, Linux, GitHub CI / CD, JUnit Testing, VSCode, IntelliJ IDEA, RStudio, Jupyter

TECHNICAL EXPERIENCE

HYPED - Hyperloop Edinburgh | Software Engineer **Edinburgh, United Kingdom**

Software Engineer (Sensors) Sep 2024 – Jun 2025

- Reworked several hundred lines of legacy C sensor code into Rust, using datasheets to implement behaviour to specification for I2C and GPIO sensors.
- Collaborated with other software teams to integrate sensor code into the wider software pipeline, ensuring sufficient error handling and extensive testing was in-place to prevent critical failures during runtime.

George Watson's College IT Team | Intern **Edinburgh, United Kingdom**

IT Intern Jun 2023 – Aug 2023

- Upgraded 300+ machines' operating systems and software to stable builds, ensuring system security. Automated steps of the machine setup process using scripts to streamline the mass update task.

JPMorgan Work Experience | Participant **Glasgow, United Kingdom**

Work Experience Programme Participant Jul 2022 – Jul 2022

- Worked in a team of six to build and present a full-stack eco-friendly initiative application with the use of React for the front-end and Express and MySQL for the back-end logic and data storage.

PROJECTS

Cache Simulator

Low-level, C, Simulation Nov 2025 – Nov 2025

- Built a configurable 2-level cache simulator with support for write-back/write-allocate policies, LRU eviction and configurable block size, capacity and associativity.
- Managed dynamic memory allocation and freeing in C, preventing memory leaks from occurring.

Scientific simulations in Python

Python, OOP, Testing, Jan 2026 – May 2026

- Developed simulations and visualisations for radioactive decay, the Mandelbrot set and cellular automata by applying OOP design features like inheritance and modularity as well as test-driven development in Python.
- Written a science journal style report on a project exploring the application of the Chaos Game algorithm in visualising frequenices of DNA nucleotide bases present in the Human Y chromosome.

Data Science Report on the Pandemic's effects on Energy Demand Peaks

Python, Pandas, Matplotlib, Statistics Feb 2026 – Apr 2026

- Analysed changes in behaviour of the daily UK electricity demand curve during the COVID-19 pandemic through a data science pipeline consisting of cleaning, exploration, visualisation, hypothesis testing and bootstrapping to determine statistical significance of any changes.

LEADERSHIP AND ACTIVITIES

CompSoc Technology Society **Edinburgh, United Kingdom**

President Jun 2025 – Jun 2026

- Led a 15+ member committee to run the biggest student tech society in Scotland, organising large-scale events for students and industry partners numbering 100+ participants and secured over £45k in sponsorship funds.
- Organised HackTheBurgh and openEuler hackathons, coordinating hackathon scheduling and technical resource provisioning of GPUs for the openEuler hackathon.

BitSIG - Low Level Programming & Hardware Special Interest Group **Edinburgh, United Kingdom**

President Jun 2026 – Present

- Run weekly low-level programming and hardware sessions for ~15 members, focusing on operating systems, CPU design and steering towards HDLs and FPGA development.

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